DD-Sudoku transformer: where the ceiling actually is

Headline: three independent capacity levers (depth/FF, head count, head width)
and a 3x training budget ALL leave end-to-end solve_rate pinned at 0.72-0.85.

New, and it reframes the goal: at the model's current N=24 the DD cascade runs
only ~1.7 rounds and full recovery sets in only above rho~0.7 (the trivial,
nearly-fully-hinted corner). There is no hint-density phase transition at N=24
to find -- the transformer has to scale to much larger N before the question
we care about is even well posed.

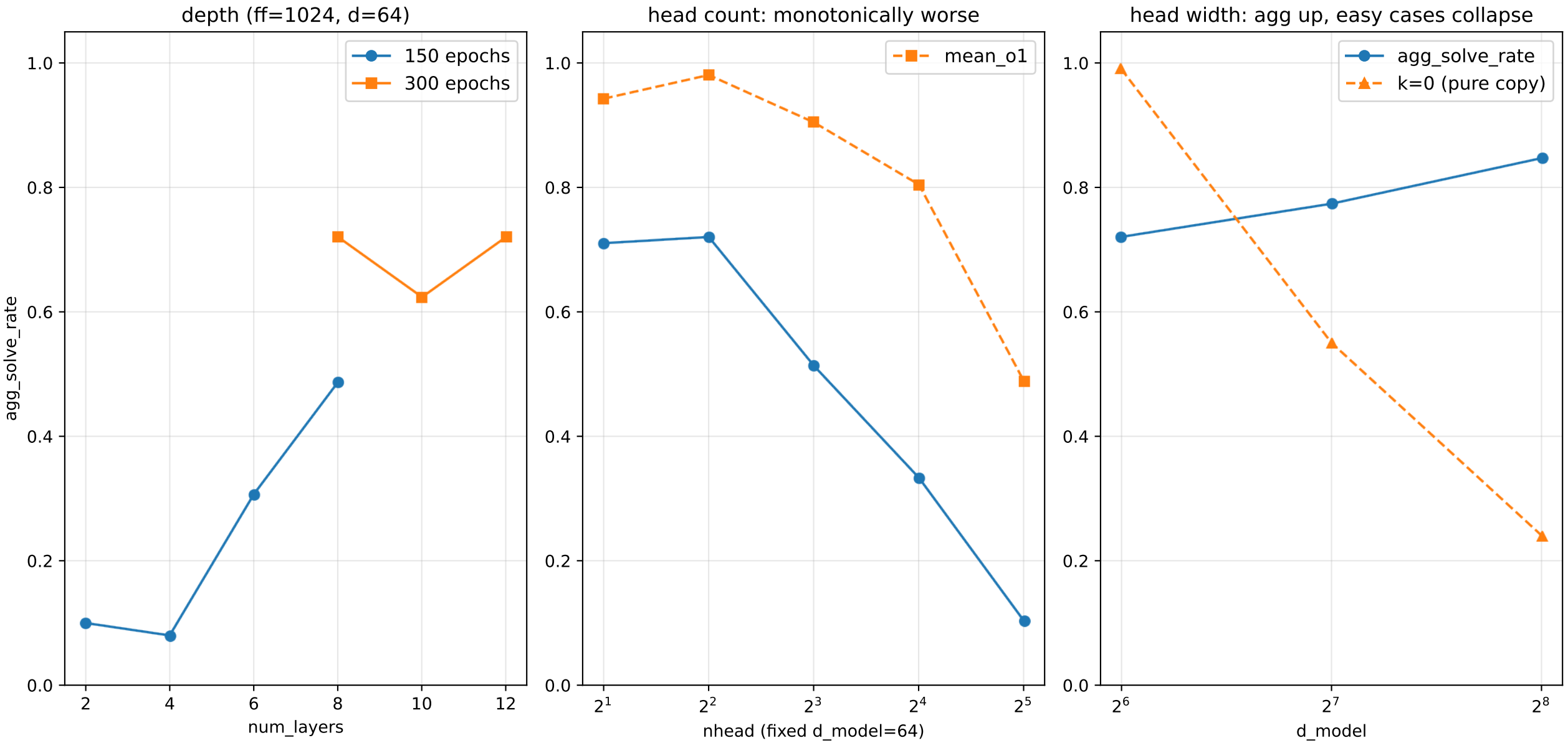
```
  branch sudoku-dd @ db6cf0a   |   q=4, d=5, N=24, seed=0   |   2026-10-19
```

Metrics: best config per lever (same eval throughout)

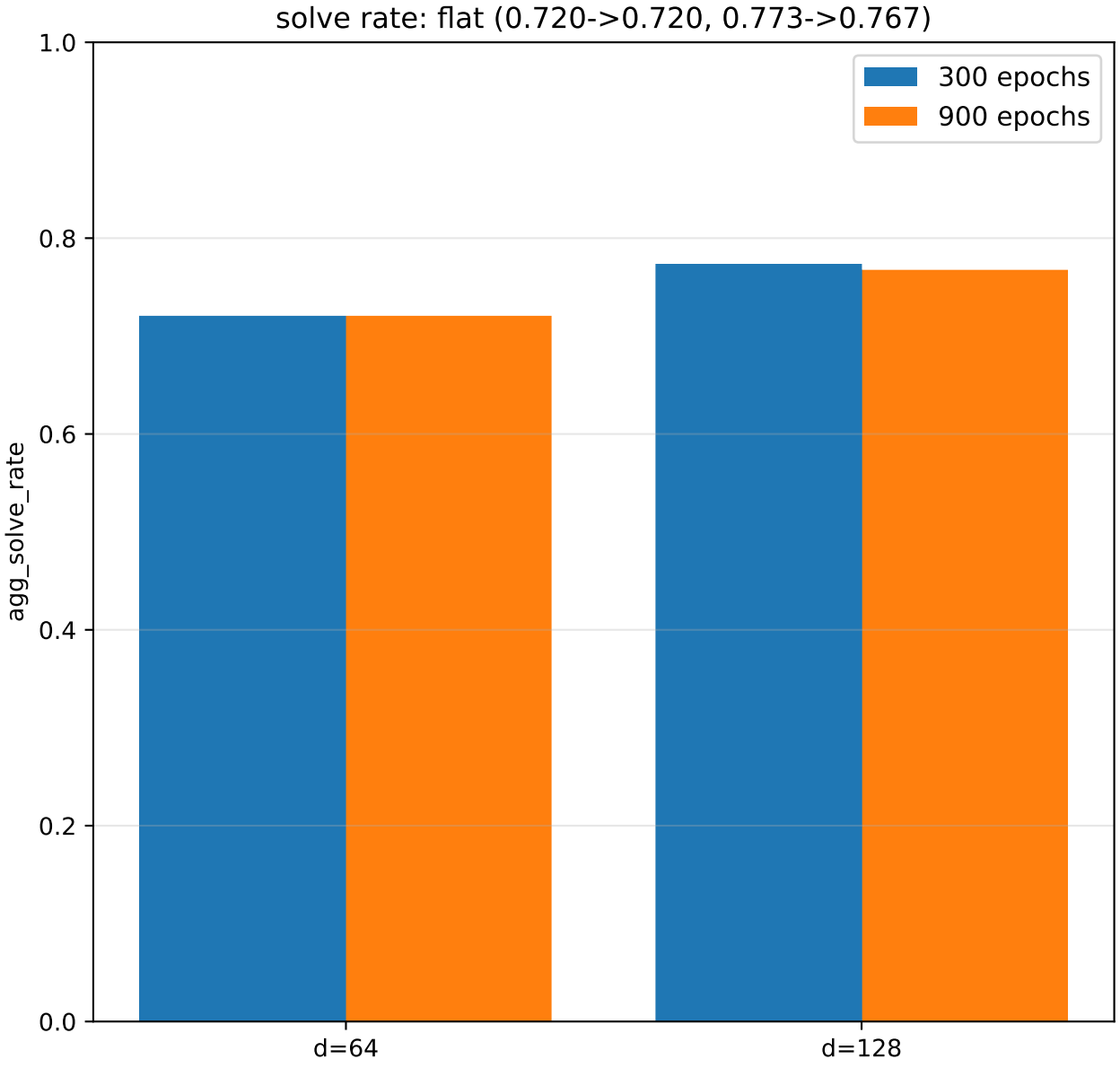
config	params	val_nll	mean_o1	k0	agg_solve_rate
L=2 ff=1024 (shallow	316,066	1.69	0.484	0.27	0.100
L=8 ff=1024 300ep	1,210,402	1.17	0.968	0.92	0.720
L=12 ff=1024 300ep d	1,806,626	1.20	0.980	0.99	0.720
+ nhead=16	1,806,626	1.29	0.804	0.76	0.333
+ d_model=128	3,994,146	1.67	0.896	0.55	0.773
+ d_model=256	9,548,834	2.41	0.842	0.24	0.847
900ep d_model=64	1,806,626	1.37	0.984	1.00	0.720
900ep d_model=128	3,994,146	2.16	0.968	0.94	0.767

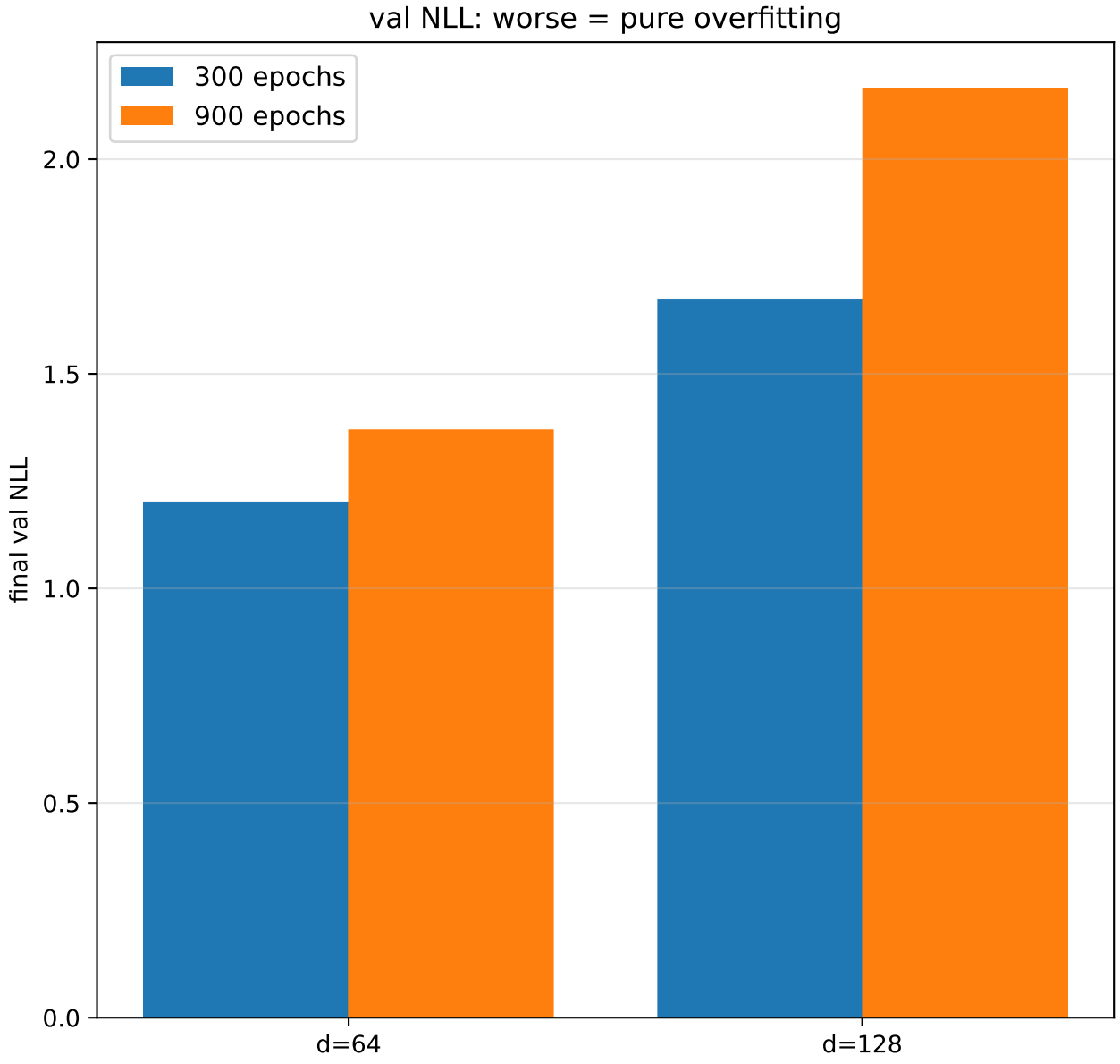
mean_o1 = solve_rate averaged over k=0..4 uncolored nodes (near-trivial end); agg = full rho in [0.3, 0.95].

The one lever that worked (depth), and two that did not

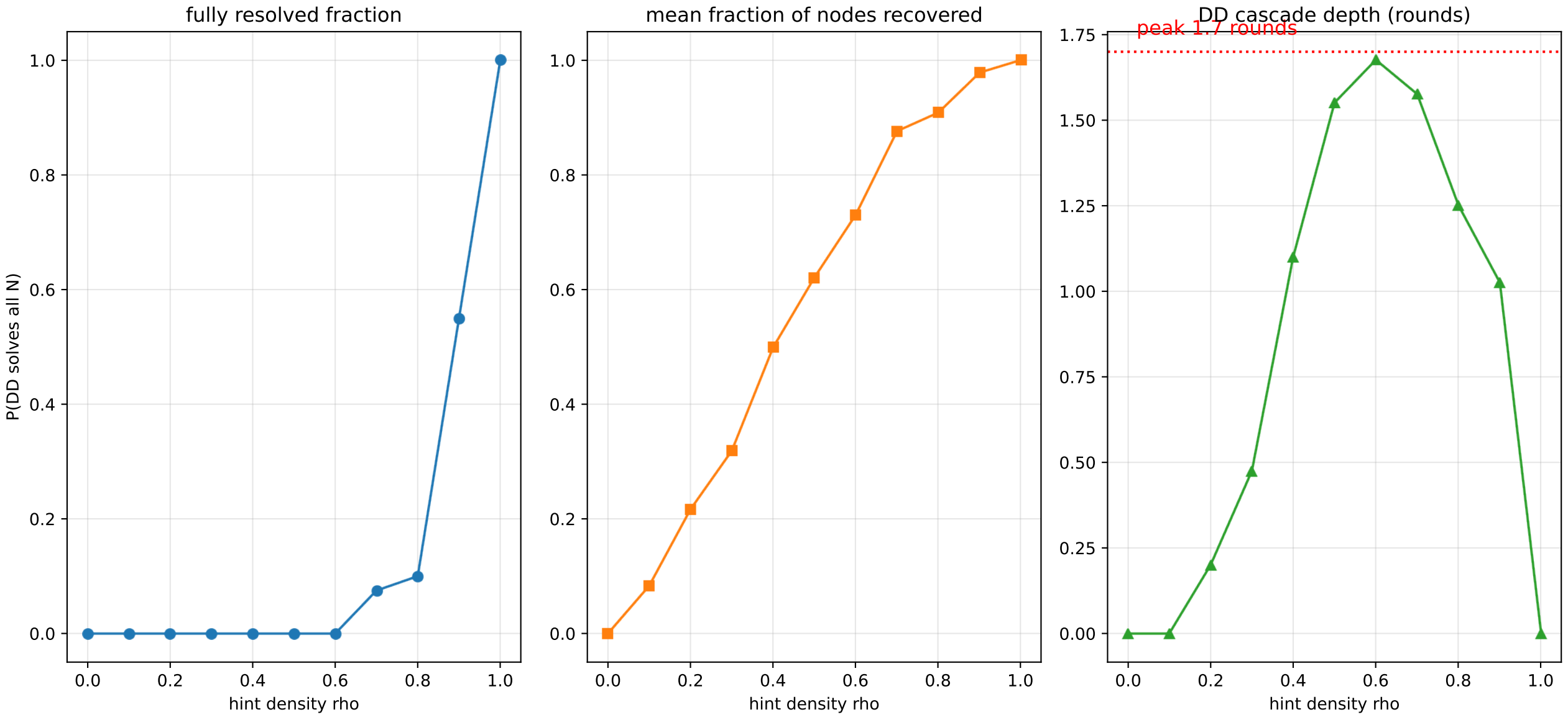


3x training budget: refuted as the explanation





Why there is no transition to find at N=24 (DD ground truth, random hints)



Pilot: 40 instances/bin, q=4 d=5 N=24. Full N-scaling sweep (N=24..384) running as Slurm 48579017.

Comparison to previous best, and what is running

Current best vs. previous best, same eval (100 instances/bin, eval seed 2e10):

```
previous  d=256 300ep : agg=0.847  mean_o1=0.842  k0=0.24
current   d=128 900ep : agg=0.767  mean_o1=0.968  k0=0.94
-> agg_solve_rate did NOT improve. The 900ep run's real contribution was
   retracting a false claim: d_model=128's bad easy-case score at 300ep was
   undertraining (mean_o1 0.896 -> 0.968), not a width penalty.
```

Running now:

```
48579017  DD reference curve, N = 24/48/96/192/384, random hints (CPU array)
48579019  scratchpad encoding: restate each deduction's evidence next to it,
          testing whether long-range retrieval is the real constraint
```

Then: scale N until the transition is visible: ~11*N tokens, so N=192 ~ 1700 tokens